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Inventor(s)	Abdelwahab; Essam Alaaeldin Ali Ahmed et al.

Apparatus for moving objects and relative movement method

Abstract

A movement apparatus for objects configured for receiving an object from an infeed station and moving the object towards an outfeed station, comprising a movement surface comprising movement cells positioned on in a matrix fashion, positioned to operate with the object, wherein each movement cell has a direction of movement which can be individually controlled; a control system configured for defining a conveying trajectory in the movement surface for conveying the object from the infeed to the outfeed station; identifying trajectory cells belonging to the conveying trajectory and directing the direction of movement of each trajectory cell to move the object in the conveying trajectory; identifying, between the movement cells, convergence cells, defining a convergence zone adjacent to the conveying trajectory and, for each convergence cell, directing the direction of movement towards a central line of the conveying trajectory to maintain of the object inside the conveying trajectory.

Inventors: Abdelwahab; Essam Alaaeldin Ali Ahmed (Samarate, IT), Filella; Carmine (Cavaria Con Premezzo, IT)

Applicant: FIVES INTRALOGISTICS S.P.A. (Lonate Pozzolo, IT)

Family ID: 1000008780423

Assignee: FIVES INTRALOGISTICS S.P.A. (Lonate Pozzolo, IT)

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Background/Summary

- (1) This invention relates to an apparatus for moving objects and a relative method for moving objects.
- (2) In particular, the invention relates to the technical sector of the systems for moving objects, for the purposes of storage, or sorting of the objects, for example logistics systems, or transport systems.
- (3) The term “object” is used to mean, in this text, a package, an item of luggage, or a bag, of any shape, weight and size, which is processed in the above-mentioned movement systems and which is therefore moved by a movement apparatus between an infeed station to an outfeed station, in order to transfer, sort it, or orient it according to a preferred direction.
- (4) Traditionally, these movement systems comprise a main conveyor, a movement apparatus, and one or more secondary conveyors, wherein the main conveyor moves the objects to be moved and/or sorted along a main path up to the movement apparatus, and the movement apparatus, to which the various secondary conveyors are connected according to predetermined criteria, moves each object to the desired secondary conveyor for transferring the object by means of the secondary conveyor to a relative distribution station.
- (5) Solutions of a different type for the movement apparatuses have been developed over the years to allow the objects to leave the main conveyor and diverge along their path towards the secondary conveyor.
- (6) One of these solutions involves the use of special carriages which move the object, typically a pack, along the main conveyor and are equipped with laterally oriented rollers which are actuated at the correct moment for directing the pack towards the desired secondary conveyor.
- (7) Another solution comprises the use of units movable above the main conveyor and configured for making contact with or engaging laterally the packs for directing them towards the secondary conveyors.
- (8) To allow a more complete and/or precise control of the position and movement parameters of the objects, movement apparatuses have recently been widespread on the market comprising a modular movement surface having a plurality of movement cells, preferably identical to each other, arranged in a matrix or chequerboard fashion on the movement surface, which are configured to operate in conjunction with each other in such a way as to move in succession objects from one, or more, infeed stations to one, or more, outfeed stations. In particular, each movement cell has a direction of movement which can be controlled independently from the other movement cells.
- (9) An example of these movement apparatuses is shown in European patent application No. EP1375389A1 which shows a transport apparatus provided with a transport surface having a plurality of conveying portions configured for making an object slide sequentially on the conveying surface. Each of these conveying portions comprises a pair of rollers parallel to each other having respective axes of rotation parallel to the conveying surface, defining a conveying direction perpendicular to the axis of rotation of the rollers. The rollers are mounted on a base, rotatable about a vertical axis relative to the conveying surface, in such a way as to vary the direction of movement of the rollers.
- (10) By means of these movement apparatuses for objects, it is advantageously possible to arbitrarily move an object on the movement surface even in the absence of mechanical components which engage on the object, to direct the object in the desired direction.

- (11) A system for controlling the movement apparatus is typically provided to define, for each object to be moved in the movement surface, a conveying trajectory from an infeed station of the object to an outfeed station of the object. For this purpose, the control system is configured for identifying, between the movement cells, trajectory cells belonging to the conveying trajectory and directing, in use, the direction of movement of each trajectory cell in the movement surface in such a way as to move the object in the conveying trajectory.
- (12) When, on the movement surface, there are two or more objects, for each one a respective conveying trajectory, an infeed station and an outfeed station is defined, in such a way that each object is moved independently from the other object, or from the other objects.
- (13) Due to the slipping between the object to be moved and the movement cells, or due to impacts between any objects present simultaneously on the movement surface, it may happen that the object loses the conveying trajectory defined for it and moves in the movement surface in an uncontrolled manner.
- (14) In this case, each object may not only not reach the relative outfeed station, but may also fall outside the movement surface, with the risk of damaging itself.
- (15) Document EP3315436 A1 relates to a device for transporting articles which is able to perform a transfer operation and a rotation operation. The conveying device has a transfer part having a belt to allow the positioning of an article, and a turntable which supports the transfer part. The belt is a continuous element suspended between a pair of rollers which slide being engaged by a transfer motor whilst a rotation motor engages with the turntable.
- (16) In this context, the technical purpose which forms the basis of this invention is to provide an apparatus for moving objects, and the relative movement method, which overcomes at least some of the above-mentioned drawbacks of the prior art.
- (17) A further aim of the invention is to provide an apparatus for moving objects, and the relative movement method, comprising a modular movement surface having a plurality of movement cells, preferably identical to each other, arranged in a matrix or chequerboard fashion on the movement surface, configured to operate in conjunction with each other in such a way as to move in succession at least one object from an infeed station to an outfeed station, which promotes a maintaining of the object inside a conveying trajectory defined for the object.
- (18) Another aim of the invention is to provide an apparatus for moving objects, and the relative movement method, which is able to increase a reliability of movement of each object inside the conveying trajectory.
- (19) The technical purpose indicated and the aims specified are substantially achieved by an apparatus for moving objects and by the relative movement method, comprising the technical features described in one or more of the appended claims.
- (20) The dependent claims correspond to possible embodiments of the invention.
- (21) Further features and advantages of the invention are more apparent in the non-limiting description which follows of a preferred non-limiting embodiment of a movement apparatus for objects and a relative method for moving objects.

Description

- (1) The description is set out below with reference to the accompanying drawings which are provided solely for purposes of illustration without restricting the scope of the invention and in which:
- (2) FIG. 1 shows a perspective view of an apparatus for moving objects made according to the invention, comprising a movement surface provided with a plurality of movement cells positioned on it in a matrix or chequerboard fashion, configured and arranged to operate in conjunction with each other and move in succession an object;

(3) FIG. 2 is an enlarged view of a detail shown in FIG. 1;

(4) FIG. 3 shows a schematic view of the movement apparatus of FIG. 1, in use, when on the movement apparatus there is an object moved from an infeed station to an outfeed station on a conveying trajectory defined on the movement surface, wherein the object is shown in successive positions on the conveying trajectory and there is an action for disturbing the object such as to make it come out from the conveying trajectory.

(5) With reference to the accompanying drawings, the numeral **1** generically denotes an apparatus for moving objects **2**.

(6) The movement apparatus **1** is configured to receive at least one object **2** from one, or more, infeed stations **3** and move the object **2** towards one, or more, outfeed stations **4**.

(7) It should be noted that FIG. 3 shows, purely by way of example, two infeed stations, **301** and **302** and two outfeed stations **401**, and **402** of the plurality of infeed stations **3**, and possible outfeed stations **4**.

(8) Preferably, the movement apparatus **1** is configured to receive several objects to be moved simultaneously from one or more of the infeed stations **3** to one or more of the outfeed stations **4** but, for simplicity, the presence of a single object in the movement apparatus **1** will be considered below.

(9) The movement apparatus **1** comprises a movement surface **101** comprising a plurality of movement cells **102** positioned on it in a matrix or chequerboard fashion, which are configured and positioned to operate in conjunction with each other and move the object **2** in succession.

(10) The object **2** is placed on the movement surface **101** in such a way that a relative lower surface (not illustrated) is resting on it.

(11) The movement apparatus **1** comprises a frame **103**, provided with respective feet **103a** configured to allow the movement apparatus **1** to be placed on the floor (not illustrated) of a movement system of the logistics, or transport, sector, in which the movement apparatus **1** can be installed. The movement apparatus **1** is shown in its entirety in FIG. 1.

(12) The frame **103** comprises a supporting mask **104**, provided with holes **104a**, and is configured to support each movement cell **102** and to ensure that an element for moving the cell, shaped like a ball **102a**, can come out from the respective hole **104a**, to allow the movement of the object **2**, as shown in more detail in FIG. 2.

(13) In fact, each movement cell **102** comprises a tubular supporting body (not illustrated) having an axis Z, which is positioned vertically, at the end of which is rotatably fixed the ball **102a**.

(14) The ball **102a** is rotatable about a horizontal axis, not illustrated, parallel to the movement surface **101** and perpendicular to the vertical axis, which, in use, defines a direction of movement D of the movement cell **102**.

(15) It should be noted that the ball **102a** is rotatable in a clockwise or anticlockwise direction, to allow a different direction of movement D' to be obtained, modified by 180° relative to the direction of movement D, that is to say, a direction of movement in the opposite direction.

(16) In detail, the direction of movement D of each movement cell **102** is perpendicular to the horizontal axis about which the ball **102a**, in use, rotates.

(17) The ball **102a** is also rotatable relative to the vertical axis Z of the supporting body, so as to be able to vary the direction of movement D of the movement cell **102**. By way of example, FIG. 2 shows a further direction of movement D1, and another direction of movement D2, as well as the corresponding directions of movement D1' and D2' in the opposite direction.

(18) For this reason, each movement cell **102** may have, in use, a respective direction of movement D and, in the direction of movement D, a respective movement speed.

(19) The movement speed derives directly from the speed of rotation of the ball **102a** about the horizontal axis.

(20) Each movement cell **102** is individually controllable, in the sense that the direction of movement D and the relative movement speed of each movement cell **102** can be controlled

independently from the other movement cells **102** forming part of the movement surface **101**.

(21) In fact, each movement cell **102** comprises a pair of actuators, not illustrated, which are independent of each other and independently controllable, which respectively control the rotation of the ball **102a** about the vertical axis Z (and that is, the direction of movement of the movement cell **102**) and the rotation of the ball **102a** about the horizontal axis (and that is, the movement speed of the movement cell **102**), as schematically shown in FIG. 2.

(22) A movement cell **102** and a movement surface **101** as described here are illustrated in detail in European patent application No. EP3643644 of the same Applicant, to which reference is made without limiting the scope of the invention.

(23) It should be noted, however, that the object of the invention, as described here, is also applicable to a movement surface **101** provided with movement cells of a different type, for example having two movement rollers, as described in the European patent application EP1375389A1 mentioned above as forming part of the prior art since, also in that case, each movement cell has a direction of movement and a respective movement speed which can be individually controlled.

(24) As mentioned above, said movement cells **102** are configured and positioned to operate in conjunction with each other and move an object **2** in succession.

(25) In fact, since the movement cells **102** are individually controllable, when the object **2** is rested on the movement surface **101**, the direction and movement speed of each movement cell **102** may be coordinated with the direction and speed of the movement cells **102** adjacent to it to ensure that the object **2** is transferred to a predetermined position of the movement surface **101**.

(26) In this way, advantageously, the movement surface **101** may be suitable for objects and packs of different dimensions, weights and features, to perform movements of different types, as described below.

(27) The movement apparatus **1** comprises a control system (not illustrated) configured to define a conveying trajectory in the movement surface **101**, for moving the object **2** from an infeed station **3** to an outfeed station **4**.

(28) It should be noted that, in general, the control system may be divided into separate functional modules, which may correspond to hardware units and/or software routines, and may consist of a single electronic device, suitably programmed to perform the functions described, for example a control PLC, or a control PC. In this case, the various functional modules can form part of the programmed device. Alternatively or in addition, the control system can comprise a plurality of electronic devices on which the above-mentioned functional modules can be distributed. In fact, the control system may have one or more processors for the execution of the functional modules, which may be distributed on different control PLCs, or different control PCs, locally, or remotely (for example, in the cloud), on the basis of the architecture of the communication network on which they reside.

(29) The term “control system” is used to comprise all the components necessary for implementing the control system, for example the electronic devices and/or hardware units and/or software routines and/or the memory modules and/or the communication network mentioned above.

(30) The control system is also connected to each movement cell **102**, for controlling the pair of actuators defining the direction of movement D and the movement speed of each movement cell **102**.

(31) As shown in FIG. 3, the movement apparatus **1** may have the two infeed stations **301** and **302** and the two outfeed stations **401**, and **402**.

(32) It should be noted that the infeed stations **3** and the outfeed stations **4** may be perimetrically distributed on a movement surface **101** and, usually, considering a longitudinal axis of extension Y of the movement surface, are positioned on an infeed side **101a** and an outfeed side **101b**, which are facing each other.

(33) In the case illustrated, the movement surface **101** has a rectangular shape and, therefore, the

infeed side **101a** and the outfeed side **101b** are short sides of the rectangle, facing each other.

(34) However, it is not necessary for the movement surface **101** to be rectangular because it may also have a different shape relative to the need for positioning the infeed stations **3**, and/or the outfeed stations **4** in the system in which the movement apparatus **1** is to be positioned.

(35) In fact, the same perimeter position of each infeed station **3**, and of each outfeed station **4**, relates to the specific need of the system.

(36) For example, the conveying trajectory may be linear, for linearly moving the object **2** between an infeed station **3** and an outfeed station **4**, positioned aligned with each other.

(37) The conveying trajectory may also be crossed, as shown in FIG. **3**, for moving the object diagonally between an infeed station **3**, for example the first infeed station **301**, to an outfeed station **4**, for example the second outfeed station **402**, wherein the first infeed station **301** and the second outfeed station **402** are offset from each other.

(38) Objects **2** different from each other, received from the movement surface **101**, can thus be sorted and/or separated to respective outfeed stations **4**, in relation to the conveying trajectory set for each object **2**. Moreover, each object **2** may also be rotated clockwise or anticlockwise about the vertical axis of the object **2**, for varying the orientation of the object **2** moved before the outfeed station **4**, if necessary.

(39) It should be noted, therefore, that the movement apparatus **1** may perform a sorting function, if it is necessary to separate the objects **2**, directing each towards specific outfeed stations **4** in relation to the properties of the object **2**.

(40) The control system is configured for identifying, between the movement cells **102**, trajectory cells **105** belonging to the conveying trajectory and directing, in use, the direction of movement **D** of each trajectory cell **105** in the movement surface **101** in such a way as to move the object **1** in the conveying trajectory.

(41) When, in use, the object **2** is in the conveying trajectory it is moved from the infeed station **3** to the outfeed station **4** in a conveying direction.

(42) The trajectory cells **105** are, therefore, among all the movement cells **102** forming part of the movement surface **101**, only those which act in conjunction with each other for moving the object **2** between the infeed station **3** and the outfeed station **4** established for the object **2**, and identify in the movement surface **101** an area, not illustrated, intended for resting the lower surface of the object **2**.

(43) According to the invention, the control system is also configured for identifying, between the movement cells **102**, convergence cells **106**, defining at least one convergence zone **107** adjacent to the conveying trajectory.

(44) The control system is configured to direct, in use, the direction of movement of each convergence cell **106**, towards a central line **T** of the conveying trajectory to promote the maintaining of the object **2** inside the conveying trajectory.

(45) It should be noted that in FIG. **3**, which schematically represents the movement surface **101**, each arrow represents the direction of movement **D** of a respective movement cell **102**, or of a respective trajectory cell **105**, or of a respective convergence cell **106** and that the conveying trajectory is the area between two lines, for example curved, in which the object **2** is intended to move. The central line **T** of the conveying trajectory is also a curved line, which represents a line of symmetry of the conveying trajectory.

(46) It should be noted that, in FIG. **3**, the numeral **102** denotes the movement cells which the control system has not defined either as trajectory cells **105** or convergence cells **106**, since they do not participate in the movement of the object **2**. The arrows indicating the movement cells **102** are illustrated in FIG. **3** with a continuous line.

(47) The arrows indicating the trajectory cells **105** are shown with dashed lines, with a non-uniform type of dashed line, whilst the arrows indicating the convergence cells **106** are shown with dashed lines, with a uniform type of dashed line.

(48) Thanks to the presence of convergence cells **106** in a convergence zone **107**, positioned adjacent to the conveying trajectory, it is possible to correct a coming out of the object **2** from the conveying trajectory.

(49) In fact, if, in use, the object **2** veers from the conveying trajectory and slides to its outside, as shown in FIG. **3** which shows a possible position for a veered object **2'**, the convergence cells **106** of the convergence zone **107** again direct the veered object **2'** towards the conveying trajectory, thus annulling the veering.

(50) The object **2**, which falls within the conveying trajectory, may therefore reach the predetermined outfeed station **4**, without the risk of reaching an incorrect outfeed station **4**, or coming out and falling from the movement surface **101**.

(51) This makes it possible to have movement apparatuses **1** which are substantially immune to the disturbances, wherein even at a maximum movement speed which can be set up for the trajectory cells **105**, there is a maximum correct movement efficiency of the objects **2**. It is not, therefore, necessary to slow down the movement speed of the trajectory cells **105**, in order to guarantee that the objects **2** remain resting on them from the infeed station **3** to the outfeed station **4**.

(52) For each convergence cell **106**, the control system is configured for setting up a respective longitudinal component of the movement speed (which extends parallel to the longitudinal axis of extension Y of the movement surface **101**) which is less than or equal to, a longitudinal component of the movement speed of a corresponding trajectory cell **105**.

(53) For each convergence cell **106**, the control system is configured for setting up a respective transversal component of the movement speed, perpendicular relative to the longitudinal axis of extension Y, which is the maximum which can be reached.

(54) In fact, each convergence cell **106** can be associated with a trajectory cell **105**, which may be, for example, the one positioned along an axis, perpendicular to the longitudinal axis of extension Y and passing through the convergence cell **106**; or which may be, for example, the one located at the minimum distance on the central line T of the conveying trajectory.

(55) The maximum speed which can be reached means the maximum speed at which the convergence cell **106** may be set.

(56) In other words, each movement cell **102** has a respective movement speed in the direction of movement D, which can be controlled independently relative to the other movement cells **102** and which may be less than, or equal to, a respective maximum speed, which is the maximum speed which can be reached.

(57) The control system is configured for setting up, for each convergence cell **106**, the respective transversal component of the movement speed, extending perpendicularly to the longitudinal axis of extension Y, which is equal to the above-mentioned maximum speed.

(58) The control system can be configured to define the convergence zone **107** and a further convergence zone **107a**, also comprising convergence cells **106**, wherein the convergence zone **107** and the further convergence zone **107a** are positioned on opposite sides of the conveying trajectory.

(59) In this way, the return of the object **2** to the conveying trajectory is promoted both on one side and on the other of the conveying trajectory.

(60) According to a version not illustrated, in addition, or alternatively relative to the presence of convergence zones opposite the conveying trajectory, the control system may be configured to detect the position of the object **2** in the conveying trajectory and define a convergence area (not illustrated), downstream of the object **2**, and another convergence area (not illustrated) upstream of the object **2** in the conveying direction from the infeed station **3** to the outfeed station **4**.

(61) In the latter case, the control system may be configured for setting up a movement speed of the convergence cells **106** of the convergence area different to the movement speed of the convergence cells **106** of the other convergence area.

(62) In particular, the movement speed of the trajectory cells **105** positioned at the object **2** may be greater than the movement speed of the convergence cells **106** of the convergence area, positioned

downstream of the object, to slow down the convergence cells **106** of the convergence area.

(63) At the same time, the movement speed of the trajectory cells **105** positioned at the object **2** may be less than the movement speed of the convergence cells **106** of the other convergence area, positioned upstream of the object **2**, to accelerate the convergence cells **106** of the other convergence area.

(64) In other words, this allows the convergence cells **106** downstream of the object **2** to be slowed down, whilst the convergence cells **106** upstream of the object are accelerated, to ensure that the object is braked in the case of veering downstream of the position of the object and, on the other hand, is accelerated in the case of veering upstream of the position of the object to promote the return of the object **2** to the conveying trajectory.

(65) In order to detect the position of the object **2**, the control system may comprise an image acquisition device configured to acquire, in use, a plurality of successive images of the movement surface **101** and a processing unit configured to process the images acquired and to check the position of the object **2** in the movement surface **101** at successive points in time, when the object **2** moves in the conveying trajectory.

(66) However, the image acquisition device is not necessary.

(67) In fact, alternatively, the position of the object can be detected, in use, by the control system performing a predictive processing since the processing unit can calculate the position of the object taking into account the instant in which the object **2** is received in the infeed station **3** and the movement speed of each trajectory cell **105** on which the object **2** is subsequently resting.

(68) If the movement apparatus **1** receives two different objects **2** in the movement surface **101** from two respective infeed stations **3** and it must move them to two respective outfeed stations **4**, the control system is configured to define two different conveying trajectories in the movement surface **101**, each associated with a respective object **2** to be moved, and to identify, for each conveying trajectory, the respective trajectory cells **105** between the movement cells **102**.

(69) In the case of two objects to be moved simultaneously in the movement surface **101**, the control system will define for each conveying trajectory, at least one respective convergence zone **107** associated with it.

(70) In use, the movement method for objects **2** according to the invention comprises the steps of receiving at least one object **2** from an infeed station **3** and moving the object **2** towards an outfeed station **4**. The movement method also comprises the steps of: preparing a movement surface **101** comprising a plurality of movement cells **102** positioned on it in a matrix or chequerboard fashion; controlling a direction of movement **D** of each movement cell **102** independently relative to the other movement cells **102**, and operating the movement cells **102** in conjunction for moving the object **2** in succession; defining a conveying trajectory in the movement surface **101** for conveying the object **2** from the infeed station **3** to the outfeed station **4**.

(71) Thanks to the control of each direction of movement and the cooperation between the movement cells **102**, it is possible to transfer the object **2** to a predetermined position of the movement surface **101**, irrespective of the dimensions, weight and features of the object **2**, thereby guaranteeing a high flexibility in the possible movements in the movement surface.

(72) The method also comprises the step of identifying, between the movement cells **102**, trajectory cells **105** belonging to the conveying trajectory and directing a direction of movement **D** of each trajectory cell **105** in the movement surface **101** in such a way as to move the object **2** in the conveying trajectory.

(73) In other words, the method comprises defining the conveying trajectory for the object **2** from the infeed station **3** to the outfeed station **4** and identifying, between all the movement cells **102** of the movement surface **101**, those trajectory cells **105** which are involved in the conveying trajectory and for which it is necessary to suitably direct the direction of movement **D** so that the object **2** is moved in the conveying trajectory.

(74) The movement method advantageously comprises the step of identifying, between the

movement cells **102**, convergence cells **106**, defining at least one convergence zone **107** adjacent to the conveying trajectory and, for each convergence cell **106**, directing the direction of movement D towards a central line T of the conveying trajectory to promote the maintaining of the object **2** inside the conveying trajectory.

(75) Thanks to the identification of at least one convergence zone **107**, comprising convergence cells **106**, any uncontrolled deviations of the object **2** from the conveying trajectory are corrected for diverting it again so as to make it fall within the conveying trajectory.

(76) It should be noted that defining both the convergence zone **107** and the conveying trajectory, before the object **2** is received in the infeed station **3**, or also when the object **2** is already present in the movement surface **101**, a condition of correction of the possible positioning of the object **2** in the movement surface **101** is always made active, which guarantees a high correction efficiency, if the object **2** were to veer.

(77) The movement further comprises the step of controlling a movement speed in the direction of movement of each movement cell **102** independently from the others, and setting up, for each convergence cell **106**, a respective longitudinal component of the movement speed, parallel to a respective longitudinal axis of extension Y of the movement surface **101**, having a value less than, or equal to, a longitudinal component of a corresponding trajectory cell **105**.

(78) In fact, the movement method further comprises the step of associating with each convergence cell **106** at least one corresponding trajectory cell **105**. The trajectory cell **105** associated with the convergence cell **106** may be that positioned, for example, along a transversal axis, perpendicular to the axis of longitudinal extension Y of the movement surface **101** and passing through the convergence cell **106**.

(79) Thanks to this association, an object **2'** which has veered and which rests on the convergence cells **106**, is moved in the conveying direction towards the outfeed station **4** with the same longitudinal speed as it would have had if it had been in the conveying trajectory.

(80) The movement method also comprises the step of setting up, for each convergence cell **106**, a respective transversal component of the movement speed, perpendicular to the longitudinal axis of extension Y, which is equal to the maximum speed which can be reached.

(81) In other words, the method comprises the step of controlling the movement speed in the direction of movement of each movement cell **102** independently from the other movement cells **102**, the movement speed of each movement cell **102** being less than, or equal to, a respective maximum speed, which is the maximum speed which can be reached; and also comprises the step of setting up, for each convergence cell **106**, a respective transversal component of the movement speed, perpendicular to the axis of longitudinal extension Y which is equal to the above-mentioned maximum speed.

(82) This favours the return of the veered object **2'** into the conveying trajectory at the maximum possible speed.

(83) The method also comprises the step of defining the convergence zone **107** and a further convergence zone **107a**, on opposite sides of the conveying trajectory.

(84) Alternatively, or in addition, the method comprises the step of detecting the position of the object **2** in the conveying trajectory and defining, in addition to the convergence zone **107**, a convergence area downstream of the object **2** in the conveying direction in the conveying trajectory, and another convergence area, upstream of the object **2** in the conveying direction.

(85) In fact, in addition to the convergence zone **107** initially associated with the convergence trajectory, it is possible to dynamically activate and deactivate a convergence area and another convergence area respectively downstream and upstream relative to the position of the object **2**, as detected. This allows an even better control of the correction of any veering of the object **2** from the conveying trajectory, in real time relative to the position of the object **2**.

(86) The method also comprises the step of setting up a movement speed of the convergence cells **106** of the convergence area different from the movement speed of the convergence cells **106** of the

other convergence area.

(87) In particular, the method comprises the step of setting up the movement speed of the trajectory cells **105** at the object **2** which is greater than the movement speed of the convergence cells **106** of the convergence area and less than the movement speed of the convergence cells **106** of the other convergence area, to slow down the convergence cells **106** of the convergence area, downstream of the object **2**, and accelerate the convergence cells **106** of the other convergence area, upstream of the object **2**.

(88) Dynamically, therefore, by defining the convergence area and the other convergence area in relation to the position of the object **2**, the movement of the object **2** on the movement surface **101** may be, advantageously, further controlled.

(89) It should be noted, therefore, that thanks to the movement apparatus **1** and the movement method according to the invention the maintaining of an object **2** inside the conveying trajectory is promoted, and the reliability of movement of each object **2** inside the conveying trajectory is improved without the need to slow down the objects **2**, even in the presence of crossed and non-linear conveying trajectories.

Claims

1. A method for moving objects (**2**) comprising the steps of receiving at least one object (**2**) from an infeed station (**3**) and moving the object (**2**) towards an outfeed station (**4**), further comprising the steps of: preparing a movement surface (**101**) comprising a plurality of movement cells (**102**) positioned on it in a matrix or chequerboard fashion; controlling a direction of movement of each movement cell (**102**) independently relative to the other movement cells (**102**), and operating the movement cells (**102**) in conjunction for moving the object (**2**) in succession; defining a conveying trajectory in the movement surface (**101**) for conveying the object (**2**) from the infeed station (**3**) to the outfeed station (**4**); identifying, between the movement cells (**102**), trajectory cells (**105**) belonging to the conveying trajectory and directing a direction of movement (D) of each trajectory cell (**105**) in the movement surface (**101**) in such a way as to move the object (**2**) in the conveying trajectory; identifying, between the movement cells (**102**), convergence cells (**106**), defining at least one convergence zone (**107**) adjacent to the conveying trajectory and, for each convergence cell (**106**), directing the direction of movement (D) towards a central line (T) of the conveying trajectory to promote the maintaining of the object (**2**) inside the conveying trajectory; controlling a movement speed in the direction of movement of each movement cell (**102**) independently from the other movement cells (**102**); setting up, for each convergence cell (**106**), a respective longitudinal component of the movement speed, parallel to a respective longitudinal axis of extension (Y) of the movement surface (**101**), having a value less than, or equal to, a longitudinal component of a corresponding trajectory cell (**105**).

2. The method according to claim 1, and comprising the step of associating with each convergence cell (**106**) at least one corresponding trajectory cell (**105**).

3. The method according to claim 1, and comprising the step of controlling a movement speed in the direction of movement of each movement cell (**102**) independently from the other movement cells (**102**), the movement speed of each movement cell (**102**) being less than, or equal to, a respective maximum speed; and further comprising the step of setting up, for each convergence cell (**106**), a respective transversal component of the movement speed, perpendicular to a respective longitudinal axis of extension (Z) of the movement surface (**101**), which is equal to the above-mentioned maximum speed.

4. The method according to claim 1, and comprising the step of defining the convergence zone (**107**) and a further convergence zone (**107a**), comprising respective convergence cells (**106**), on opposite sides of the conveying trajectory.

5. The method according to claim 1, and comprising the step of detecting the position of the object

(2) in the conveying trajectory and defining a convergence area, downstream of the object (2) in a conveying direction in the conveying trajectory from the infeed station (3) to the outfeed station (4), and another convergence area, upstream of the object (2).

6. The method according to claim 5, and comprising the step of controlling a movement speed in the direction of movement of each movement cell (102) independently from the others; setting up a movement speed of the convergence cells (106) of the convergence area different to the movement speed of the convergence cells (106) of the other convergence area.

7. The method according to claim 6, and comprising the step of setting up the movement speed of the trajectory cells (105) at the object (2) which is greater than the movement speed of the convergence cells (106) of the convergence area and less than the movement speed of the convergence cells (106) of the other convergence area, to slow down the convergence cells (106) of the convergence area and accelerate the convergence cells of the other convergence zone.
